

Act-0 Results Analysis

Reproducing EEG-VJEPA on TUAB, our FEI+C model, and healthy-vs-stroke classification: eight experiments and a visual check explained

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Results at a glance

The story in three sentences.

1. We reproduced the EEG-VJEPA paper's TUAB evaluation faithfully, but the released checkpoints fall far short of the paper, because their representation is **collapsed**: every recording gets almost the same embedding.
2. Our own model, **FEI+C**, is not collapsed. It works on NMT, where it was trained, and it transfers to TUAB, where it performs at least as well as the released ViT-M.
3. On the stroke data, the clinical baseline (BSI) is strong and stable, and FEI+C matches it.

#	Question	Answer	Key number (reference)
1	Do the released checkpoints reach the paper's frozen TUAB AUROC?	No , and they lose to random weights	ViT-M 0.671 ± 0.020 (paper 0.877; random ViT-M 0.787 ± 0.006)
2	Does fine-tuning the released ViT-M beat training from scratch?	No measurable gain	0.755 ± 0.053 vs 0.768 ± 0.114 (paired $p = 0.81$; paper 0.885)
3	Why? Are the released features collapsed?	Yes , on TUAB and on stroke	token std 0.0051 vs 0.634 for random weights
4	With a simple linear probe, how far do the released features get?	Above band power, not clearly above random weights	0.838 full train (paper 0.877; band power 0.757)
5	Does our FEI+C work on NMT?	Yes ; both branches add up	FEI+C 0.865 ± 0.024 vs FEI 0.819 ± 0.029 (better in 5/5 folds)
6	Does FEI+C transfer to TUAB?	Yes : ties ViT-M, beats band power	FEI+C 0.862 vs ViT-M 0.838 ($\Delta +0.024$, $p = 0.147$)
7	Stroke: is our pipeline sound, and how strong is the clinical BSI baseline?	Yes ; BSI is strong and stable	BSI 0.907 primary / 0.938 sensitivity (LOSO; chance 0.5)
8	Stroke: with a corrected input prep, how does FEI+C do?	Matches BSI	FEI+C 0.926 vs BSI 0.907 (LOSO; chance 0.5)
V	Do t-SNE / UMAP maps of the TUAB features show normal and abnormal clusters?	No , for learned and random weights alike	10-NN label agreement 0.598 – 0.629 on the original features (chance 0.502)

Names used throughout. *ViT-M* and *ViT-B* are the two released EEG-VJEPA checkpoints. *FEI+C* is our two-branch model (FEI = frequency-masked waveform encoder, C = spectrogram-dynamics encoder), explained in the Act-0 teaching guide. *rand-X* means the same network as X with random, untrained weights.

How to read the numbers

This page explains every metric and test used later. Nothing here is a result.

AUROC (area under the ROC curve) is our main metric. Take one random abnormal (or stroke) recording and one random normal (or control) recording. AUROC is the probability that the model gives the abnormal one the higher "abnormal" score. **0.5 is chance, 1.0 is perfect**. It does not depend on any decision threshold. We write it as 0.xxx everywhere. The **ROC curve** plots the true-positive rate against the false-positive rate for every possible threshold; AUROC is the area under it.

Accuracy, balanced accuracy (BAcc), sensitivity, specificity. These need a threshold (we always use $P = 0.5$, never tuned). *Sensitivity* = fraction of abnormal/stroke recordings caught; *specificity* = fraction of normal/control recordings correctly called normal; *BAcc* = their average, which is not fooled by unequal class sizes. **AUC-PR** is the area under the precision-recall curve; its chance level equals the fraction of positives, so it is only comparable within one data set.

Mean $\pm$ sd. When an experiment is repeated on several training subsets (TUAB) or folds (NMT), we report the mean and the standard deviation across them (population sd). The sd shows how much the result depends on which data were used.

95 % confidence interval (CI). Made by the *bootstrap*: the test recordings (or subjects) are resampled with replacement 2,000 times and the metric is recomputed each time. The middle 95 % of those values is the CI. A wide CI means the number is uncertain. For a **difference** between two models (Δ AUROC), both models are scored on the same resamples; if the CI of Δ contains 0, we cannot tell the two models apart.

p-value. The probability of seeing a result at least this extreme if there were really no effect. Small p (conventionally < 0.05) means the effect is unlikely to be luck. A large p does **not** prove there is no effect; it means the data cannot show one. For Δ AUROC we report $p(\Delta \leq 0)$: the share of bootstrap resamples in which the first model is not better.

Which test where.

- *Mann-Whitney*: compares two groups of numbers without assuming a normal distribution (released runs vs random-weight runs; stroke vs control feature values).
- *Paired t-test / Wilcoxon*: compare two methods run on the **same** subsets (fine-tune vs scratch on s0–s4).
- *Permutation test*: the labels (stroke/control) are shuffled 1,000 times and the whole evaluation is re-run each time. This gives the AUROC values that pure chance produces with this exact pipeline. The p-value is the share of shuffles that did at least as well as the real labels.

How the stroke models are tested. There are only 15 subjects, so every scheme holds subjects out:

- *LOSO* (leave-one-subject-out): train on 14 subjects, predict the 15th, repeat for each subject. This was the pre-registered scheme.
- *5-fold $\times 100$* : split subjects into 5 groups, train on 4 and predict 1, and repeat the whole thing with 100 different random splits (the same 100 splits for every model). This shows how much the result depends on the split.
- *LPO* (leave-pair-out): hold out one stroke and one control subject at a time, for all pairs.

Random-init control. The most important check in this document. We run the same network with random, untrained weights through exactly the same pipeline. If the pre-trained model does not beat its random twin, the pre-training did not help: the score comes from the architecture and the classifier, not from learned knowledge.

Collapse. A known failure of self-supervised training. The encoder learns to give (almost) the same output for every input, which makes its training task trivially easy. We measure it two ways: *token std* (how much the encoder's output varies within a recording) and *mean cosine between recordings* (1.0 means every recording gets the same embedding).

Frozen, fine-tune, scratch. *Frozen*: the encoder is fixed and only a small classifier ("head" or "probe") on top is trained. *Fine-tune*: encoder and head are trained together, starting from the pre-trained weights. *Scratch*: the same, but starting from random weights. Fine-tune minus scratch = what pre-training is worth.

Final vs best epoch. The paper's recipe trains for 500 epochs with no validation set. We report the **final** epoch as our main number. We also show the *best-epoch** number: the highest test AUROC over all 500 epochs. It is marked with * because it is picked by looking at the test set, so it is optimistic and not a fair result.

Part A — Reproducing EEG-VJEPA on TUAB

Experiment 1: frozen reproduction of the paper's TUAB result

Question. With the authors' released checkpoints and their own evaluation code, do we get the paper's TUAB AUROC (ViT-M 0.877, ViT-B 0.879)? And did we reproduce their pipeline faithfully?

Setup.

Item	Value
Data	TUAB v3.0.1; test = the official eval set (276 recordings, never used for training)
Preprocessing	Paper §4.1: 19 channels, first 5 min, 1–40 Hz, 100 Hz, per-channel z-score, 5 s frames every 2.5 s
Encoders	Released ViT-M and ViT-B, frozen ; random-init ViT-M as control (3 seeds)
Heads	§4.2 head (the authors' AttentionClassifier) and Fig-3 head (AttentiveClassifier)
Training recipe	Authors' own: 500 epochs, batch 2, lr 1e-3
Training data	5 class-balanced, patient-disjoint subsets s0–s4 of 546 recordings (as in the paper)
Reported	Final-epoch AUROC, mean $\pm$ sd over subsets

Is our pipeline faithful?

- We compared our code with the authors' public repository (commit `c739fad`): **43 of 48 files are byte-identical**.
- The other 5 files (`app/main.py`, `app/vjepa/train.py`, `evals/eeg_classification_frozen/eval.py`, `src/models/attentive_pooler.py`, `src/utils/schedulers.py`) contain only our additions, all switched off by default or not affecting the result: a hardware temperature guard, a per-run random seed (upstream always uses seed 0, so its "5 runs" could not differ), balanced accuracy logged instead of AUPRC, an option to choose the head (default = the authors' head), an option for random-init weights, and a learning-rate scale that is 1 unless explicitly set (never set in Act-0).
- The checkpoints are the authors' released files: SHA-256 ViT-M `5fee4a6c08123cfb...`, ViT-B `66e9673f8ce42c3c...`.

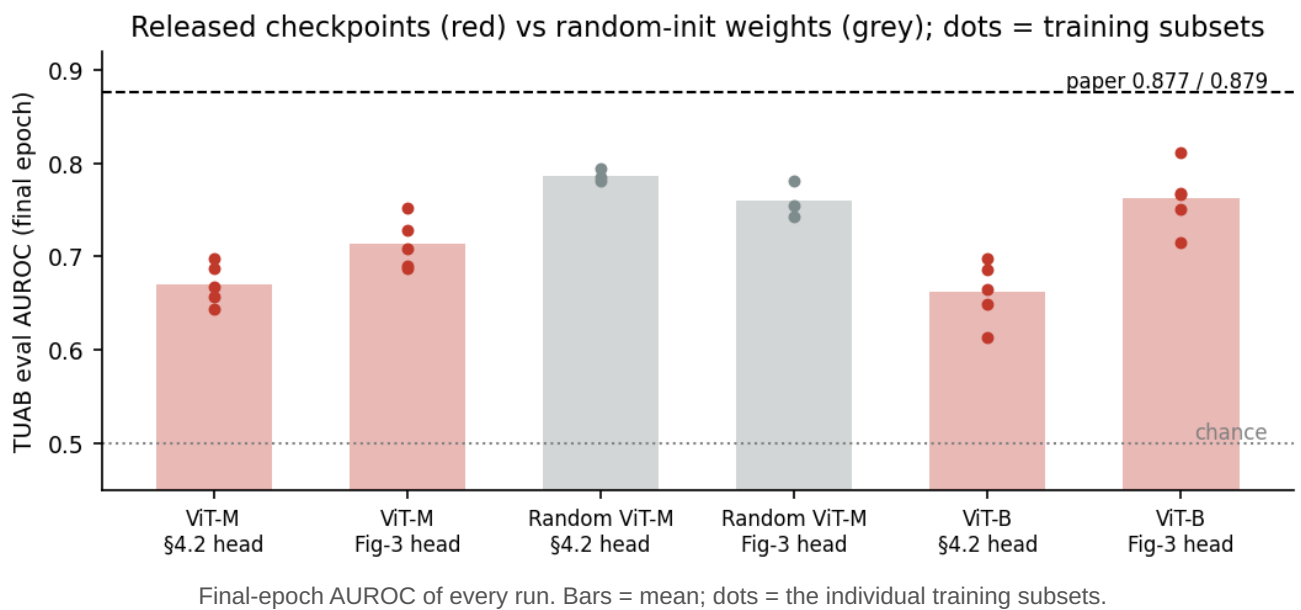
Result.

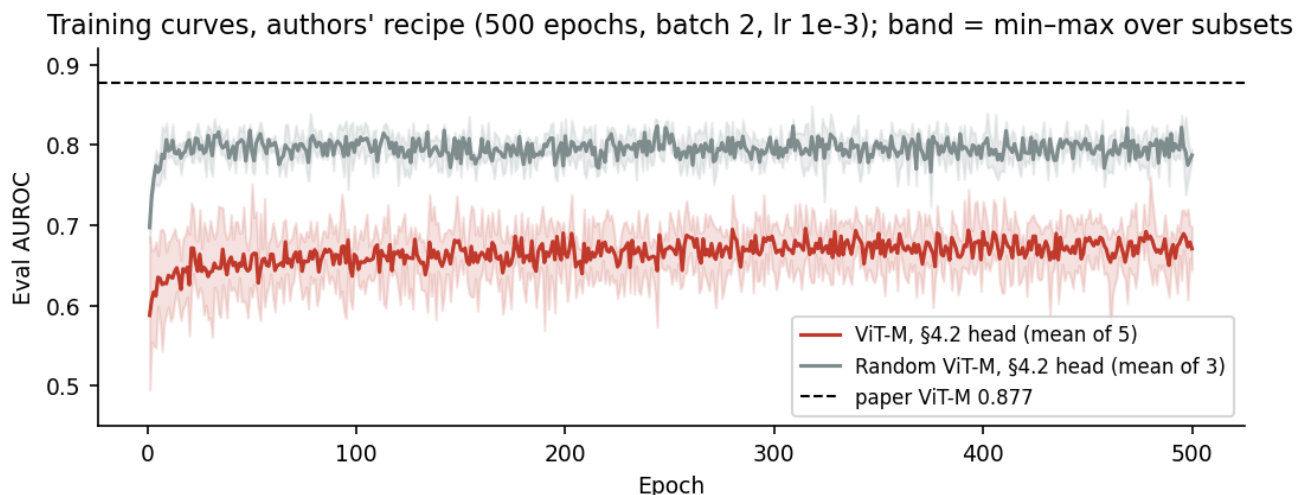
Encoder and head	Runs	Acc	F1	AUROC (final epoch)	Best-epoch AUROC*	Paper AUROC
ViT-M, §4.2 head	5	0.628 ± 0.024	0.626 ± 0.023	0.671 ± 0.020	0.737	0.877
ViT-M, Fig-3 head	5	0.664 ± 0.035	0.660 ± 0.033	0.713 ± 0.025	0.784	0.877
Random ViT-M, §4.2 head	3	0.709 ± 0.010	0.704 ± 0.006	0.787 ± 0.006	0.844	—
Random ViT-M, Fig-3 head	3	0.707 ± 0.023	0.705 ± 0.023	0.760 ± 0.016	0.839	—
ViT-B, §4.2 head	5	0.595 ± 0.021	0.594 ± 0.021	0.662 ± 0.030	0.715	0.879
ViT-B, Fig-3 head	5	0.694 ± 0.033	0.687 ± 0.030	0.762 ± 0.031	0.814	0.879

*Best-epoch AUROC is optimistic (chosen on the test set); see the primer.

Per-subset final AUROC:

Encoder and head	s0	s1	s2	s3	s4
ViT-M, §4.2 head	0.644	0.698	0.656	0.667	0.688
ViT-M, Fig-3 head	0.687	0.729	0.690	0.709	0.752
Random ViT-M, §4.2 head	0.786	0.795	0.781	—	—
Random ViT-M, Fig-3 head	0.781	0.755	0.742	—	—
ViT-B, §4.2 head	0.649	0.613	0.665	0.698	0.687
ViT-B, Fig-3 head	0.767	0.811	0.715	0.751	0.767





How to read it.

- Every released-checkpoint result is far below the paper line. ViT-M with the paper's main head reaches 0.671 ± 0.020 , which is **0.206 below** the paper's 0.877. Even the optimistic best-epoch number (0.737) stays far below.
- The grey bars are the surprise: a ViT-M with **random, untrained weights** does better (0.787 ± 0.006) than the released pre-trained one.
- The training curves show this is not a matter of training too briefly: the gap is there from the first epochs to epoch 500.

Is it real?

- Random weights beat the released ViT-M by 0.116 with the §4.2 head (Mann-Whitney $p = 0.018$) and by 0.046 with the Fig-3 head ($p = 0.036$). Both are significant.
- The two groups touch only at the edge: with the Fig-3 head, the best released run (0.752) is slightly above the worst random run (0.742).

Takeaway. The pipeline is faithful, but the released checkpoints do not reproduce the paper. They even lose to random weights, which means the pre-trained features are worse than no pre-training. Experiment 3 shows why.

Limits. The paper does not say which 546 recordings form each subset, which epoch it reports, or whether TUAB-eval was seen in pre-training; our choices were fixed in advance. Only 3 random-init seeds were run.

Experiment 2: fine-tuning vs training from scratch

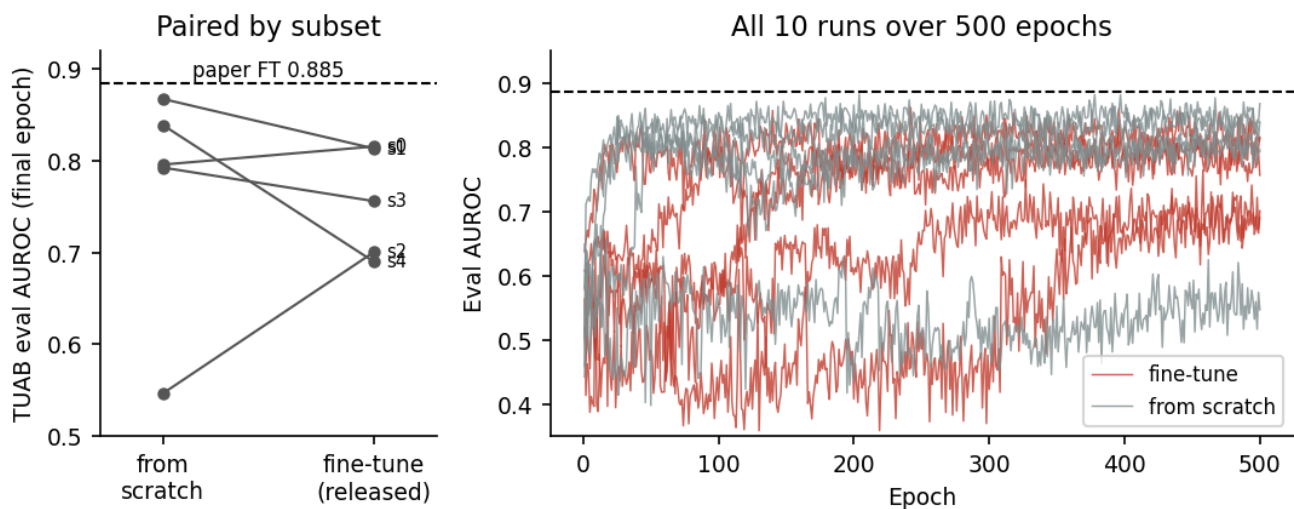
Question. The paper also reports fine-tuning the whole ViT-M (AUROC 0.885). Does starting from the released weights help compared with training the same network from random weights?

Setup.

Item	Value
Network	ViT-M + §4.2 head, all weights trained
Start	Fine-tune = released checkpoint; scratch = random weights
Recipe	Authors' fine-tune config: 500 epochs, batch 2, lr 1e-3 (cosine to 0), weight decay 1e-3
Precision	bf16 — a labelled deviation (upstream uses full precision); chosen for speed and used for all 10 runs
Data / test	Subsets s0–s4 / official eval set, paired by subset

Result.

Run	s0	s1	s2	s3	s4	Mean $\pm$ sd
Fine-tune (released start), final AUROC	0.815	0.813	0.701	0.756	0.690	0.755 $\pm$ 0.053
From scratch (random start), final AUROC	0.796	0.867	0.547	0.792	0.839	0.768 $\pm$ 0.114
Difference (fine-tune – scratch)	+0.020	-0.054	+0.154	-0.037	-0.149	-0.013
Fine-tune, best-epoch AUROC*	0.861	0.845	0.749	0.842	0.764	0.812 $\pm$ 0.046
From scratch, best-epoch AUROC*	0.848	0.882	0.629	0.854	0.868	0.816 $\pm$ 0.094
Fine-tune: epochs with every eval recording called normal	28	50	165	8	317	—
Scratch: epochs with every eval recording called normal	22	4	190	8	3	—



Left: each line joins the scratch and fine-tune result for the same subset. Right: test AUROC of all 10 runs over 500 epochs.

How to read it.

- Fine-tune (0.755 ± 0.053) and scratch (0.768 ± 0.114) are the same within noise. Fine-tuning wins on 2/5 subsets; the mean difference is -0.013.
- The right panel shows why the spread is so large: with lr 1e-3 and batch 2, several runs spend long stretches predicting "normal" for every recording (up to 317 epochs, last table row) before recovering as the learning rate decays.

Is it real? No difference can be shown: paired t-test $p = 0.81$, Wilcoxon $p = 0.81$. The best-epoch* numbers (0.812 ± 0.046 vs 0.816 ± 0.094) tell the same story.

Takeaway. Under the authors' released recipe, the released pre-training is worth nothing measurable: the network does as well from random weights. This fits the collapse found in Experiment 3.

Limits. bf16 is a deviation from the upstream config. The paper's own §5 says its fine-tuning "benefited from lower learning rates and partial unfreezing", so the recipe behind its 0.885 is probably not the released config and was never published. The very large seed spread means 5 subsets can only detect big differences.

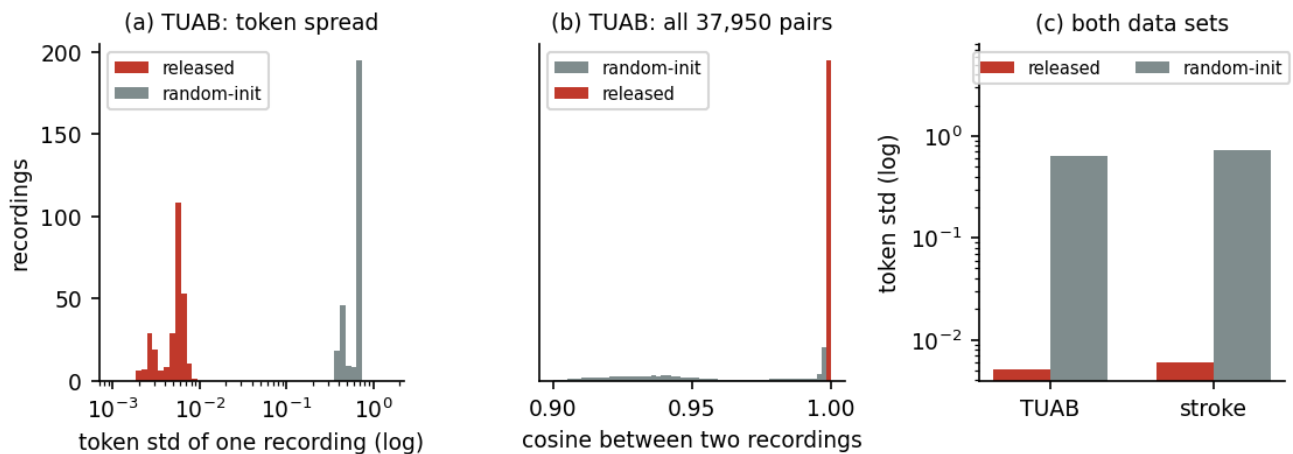
Experiment 3: the released encoder is collapsed

Question. Why do the released features lose to random ones? We test whether the encoder gives the same output for every recording (collapse).

Setup. Released ViT-M and random-init ViT-M (3 seeds), frozen, on the 276 TUAB eval recordings and the 15 primary stroke subjects (Part C). For each recording we measure the token std and the cosine similarity between every pair of recordings.

Result.

Data set	Encoder	Token std (mean)	Mean cosine between recordings	Recordings
TUAB eval	Released ViT-M	0.0051	0.999982	276
TUAB eval	Random-init ViT-M (3 seeds)	0.634	0.974	276
Stroke (primary)	Released ViT-M	0.0060	0.999994	15
Stroke (primary)	Random-init ViT-M (3 seeds)	0.717	0.9991	15



(a) Token std of every TUAB recording (log scale). (b) Cosine similarity for all pairs of TUAB recordings. (c) Mean token std on both data sets (log scale).

How to read it.

- (a) The released encoder's outputs barely vary: its token std is about **125 times smaller** than for random weights on TUAB (119 times on stroke).
- (b) Almost every pair of TUAB recordings has cosine ≈ 1 with the released encoder (mean 0.999982): normal and abnormal recordings look the same to it. Random weights keep recordings apart (mean 0.974).
- (c) The same happens on the stroke data, which the encoder never saw.

Is it real? The two groups do not overlap at all: the largest released token std of any recording is 0.0085, the smallest random one is 0.379.

Takeaway. The released checkpoints are collapsed. A classifier on top has almost no information to work with, which explains Experiments 1 and 2. This is a property of the checkpoint, not of our input: it appears on TUAB and on a completely different data set (stroke).

Limits. Collapse metrics show that the features carry little information; they cannot tell us *why* the authors' pre-training collapsed.

Experiment 4: the simplest possible probe

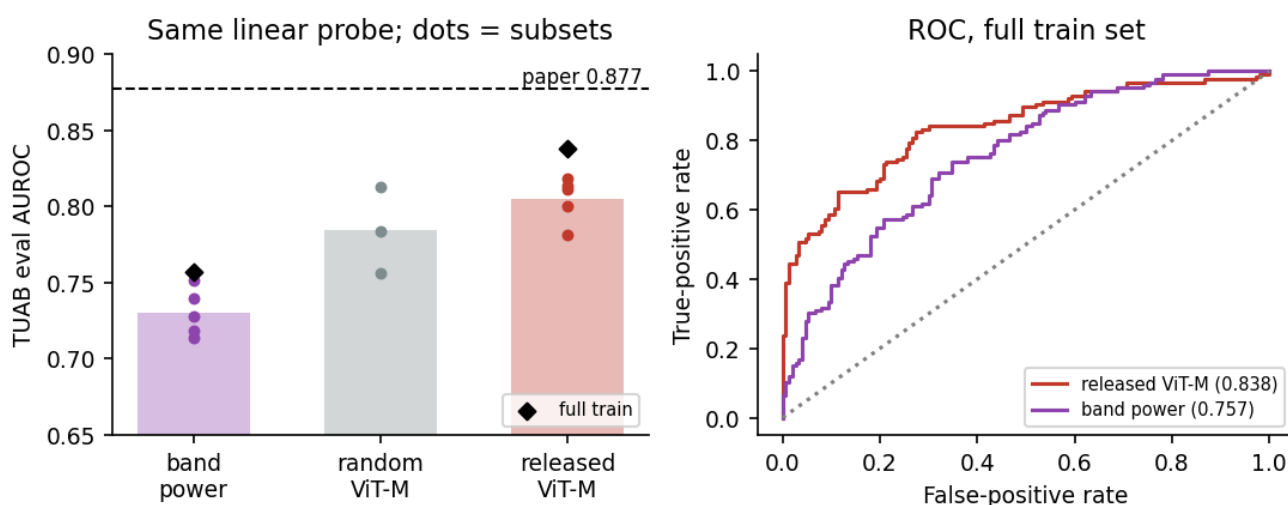
Question. If the collapsed features still contain a small signal, can a very simple classifier find it? We also add a classical, hand-crafted baseline (band power) as a reference.

Setup.

Item	Value
Features	Released ViT-M (frozen): token mean + token std, averaged over the recording (768 numbers); random-init ViT-M (3 seeds); band power (hand-crafted relative power per channel and band)
Classifier	StandardScaler + logistic regression (C = 1, balanced classes)
Training data	Subsets s0–s4, and the full TUAB training set (2,717 recordings)
Test	Official eval set (276)

Result.

Features	Subsets (546 recordings each)	Mean $\pm$ sd	Full train (2,717)
Band power (hand-crafted)	0.719 / 0.752 / 0.740 / 0.714 / 0.728	0.730 $\pm$ 0.014	0.757
Random-init ViT-M	0.813 / 0.756 / 0.784	0.784 $\pm$ 0.023	—
Released ViT-M	0.813 / 0.811 / 0.800 / 0.781 / 0.818	0.805 $\pm$ 0.013	0.838 [0.787, 0.884]



Left: AUROC per subset (dots) and on the full training set (black diamond). Right: ROC curves on the full training set.

How to read it.

- The simple probe does much better than the paper's head in Experiment 1 (0.134 higher on the same subsets). A likely reason: StandardScaler re-scales every feature to spread 1, which blows the tiny remaining differences back up.
- Trained on all 2,717 recordings, the released ViT-M reaches 0.838, the best released-checkpoint number we have. It is still **0.039 short** of the paper's 0.877.
- Both ViT-M variants beat band power.

Is it real?

- Released vs random weights on the subsets: 0.805 ± 0.013 vs 0.784 ± 0.023 , Mann-Whitney $p = 0.196$. The released weights are **not** clearly better than random ones.
- The full-train CI [0.787, 0.884] is wide enough to include the paper's 0.877, so this single number cannot rule the paper out on its own. On the paper's own protocol (the 5 subsets), every run is below 0.877 (table above).

Takeaway. Even with the most forgiving probe, the released checkpoint does not reach the paper, and it cannot be told apart from random weights. This number (0.838) is the fair ViT-M reference for our own model in Experiment 6.

Limits. The band-power feature set is simple (relative power only); a richer hand-crafted set would likely score higher. Only 3 random seeds.

Part B — Our model: FEI+C

The released EEG-VJEPA could not serve as a working pre-trained encoder, so we used our own two-branch model, FEI+C, built earlier on NMT. Part B checks that it is healthy where it was trained (NMT) and that it transfers to TUAB.

Experiment 5: FEI+C on NMT

Question. Does FEI+C, in the exact configuration used in Act-0, work on NMT? Do both branches contribute?

Setup.

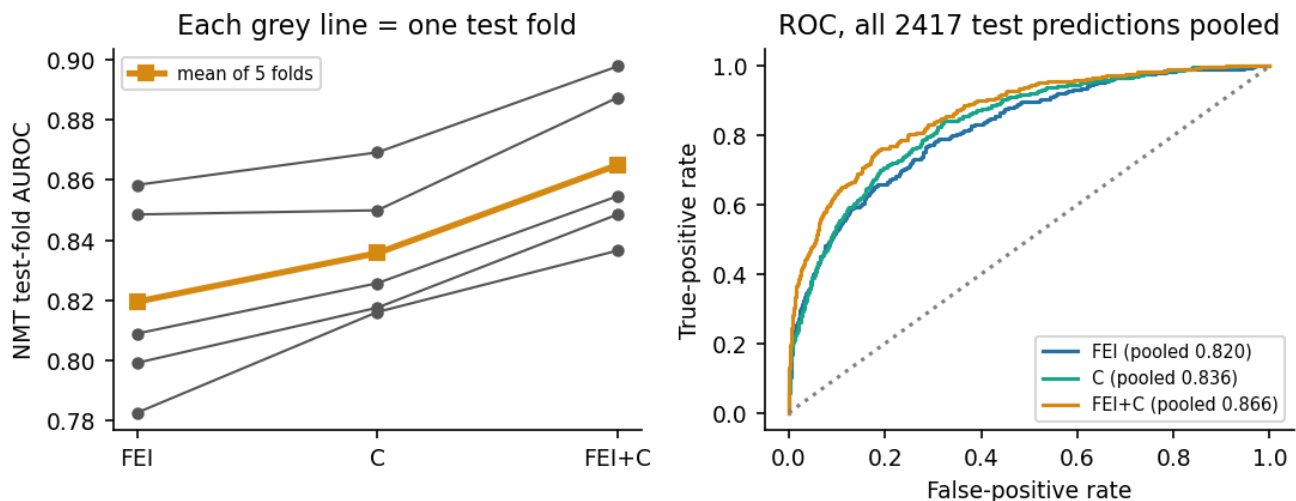
Item	Value
Data	NMT normal/abnormal, 2417 recordings (415 abnormal)
Protocol	5-fold cross-validation (seed 0), recording level; each fold's encoders were pre-trained only on that fold's training recordings
Encoders	Plain FEI (20 s windows) and Branch C (20 s windows), frozen
Classifier	StandardScaler + logistic regression (C = 1, balanced)
Arms	FEI alone, C alone, FEI+C (the two feature vectors joined)

Gate: are we loading the encoders correctly? Before computing new numbers, we re-ran the older, already-recorded FEI configuration (5 s windows) with the same loading code. It reproduces every recorded fold AUROC within 2.7e-04:

Fold	Re-computed now	Recorded in Act-1	Absolute difference
f0	0.8580	0.8583	2.7e-04
f1	0.7954	0.7953	3.0e-05
f2	0.8471	0.8472	9.0e-05
f3	0.7870	0.7872	1.2e-04
f4	0.8040	0.8040	3.0e-05

Result.

Arm	f0	f1	f2	f3	f4	AUROC mean $\pm$ sd	BAcc	AUC-PR
FEI	0.858	0.799	0.849	0.782	0.809	0.819 $\pm$ 0.029	0.729	0.563
C	0.869	0.818	0.850	0.816	0.826	0.836 $\pm$ 0.021	0.752	0.572
FEI+C	0.898	0.849	0.888	0.837	0.855	0.865 $\pm$ 0.024	0.778	0.657



Left: AUROC per test fold; each grey line is one fold. Right: ROC of all test predictions pooled.

How to read it. C alone is better than FEI alone, and joining them is better than either: every grey line rises from left to right.

Is it real? FEI+C beats FEI by +0.046 (in 5/5 folds) and beats C by +0.029 (in 5/5 folds). BAcc (0.778) and AUC-PR (0.657) improve in the same way.

Takeaway. FEI+C works on NMT, and the two branches carry different information: C adds to FEI in every fold.

Limits. No random-init control was run for this configuration on NMT, so we can say that the two branches add up, but not how much of that comes from pre-training. With 5 folds, "better in every fold" is the strongest evidence available, but it is not a formal significance test.

Experiment 6: FEI+C transferred to TUAB

Question. FEI+C was built on NMT (Pakistan). Does it transfer to TUAB (USA), a different hospital and population? How does it compare with the released ViT-M under the **same** probe, and does its pre-training beat random weights?

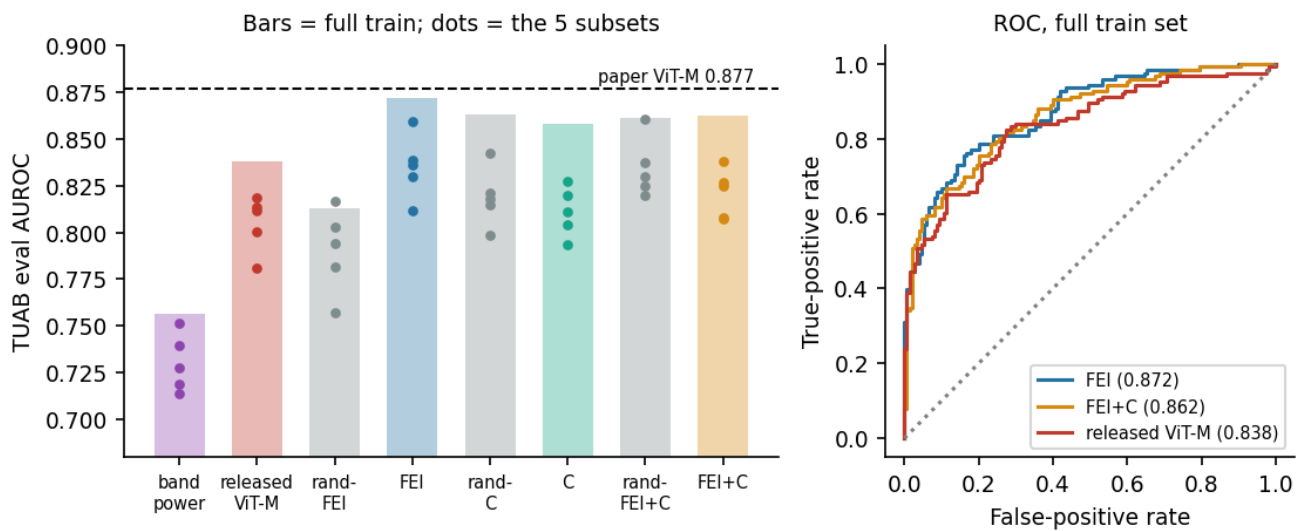
Setup.

Item	Value
Data / test	TUAB, official eval set (276), same subsets s0–s4 and full train set as Experiment 4
FEI+C input prep	19 channels, first 20 min, 0.5–40 Hz, 200 Hz, per-channel z-score
FEI	The 5 NMT-pretrained fold encoders
C and FEI+C	Encoders pre-trained on NMT plus the unlabelled TUAB training recordings ("NMT+TUAB-train pretrained"), so C sees its own input distribution. The FEI inside this FEI+C is the jointly pre-trained FEI. TUAB-eval was never used in pre-training
Controls	Random-init FEI, C and FEI+C (3 seeds)
Classifier	Same probe as Experiment 4; the 5 fold encoders' predictions are averaged
Comparisons	Paired bootstrap Δ AUROC on the 276 eval recordings

Result.

Features (all: same linear probe)	Subsets s0–s4	Full train (2,717)	Full train, single encoders
Band power (hand-crafted)	0.730 ± 0.014	0.757	—
Released ViT-M	0.805 ± 0.013	0.838	—
FEI (NMT-pretrained)	0.835 ± 0.015	0.872	0.846 ± 0.011
rand-FEI	0.790 ± 0.020	0.813	0.786 ± 0.016
C (NMT+TUAB-train pretrained)	0.811 ± 0.012	0.858	0.834 ± 0.009
rand-C	0.819 ± 0.014	0.863	0.838 ± 0.010
FEI+C (NMT+TUAB-train pretrained)	0.821 ± 0.012	0.862	0.841 ± 0.003
rand-FEI+C	0.834 ± 0.014	0.861	0.837 ± 0.002

"Full train, single encoders" = the mean $\pm$ sd of the 5 fold encoders scored one at a time, instead of averaged.



Left: bars = full training set, dots = the 5 subsets. Right: ROC on the full training set.

Are the encoders healthy? Unlike the released ViT-M, both FEI+C encoders keep recordings apart:

Encoder	Embedding std across recordings	Mean cosine between recordings
FEI (NMT-pretrained), 5 fold encoders	0.46–0.58	0.73–0.78
C (NMT+TUAB-train pretrained), 5 fold encoders	0.36–0.39	0.59–0.69
Released ViT-M (for comparison)	0.0024	0.999982

How to read it.

- Every FEI+C variant is well above band power (0.862 vs 0.757).
- FEI+C and the released ViT-M are close: 0.862 vs 0.838.
- FEI alone is the highest bar (0.872).
- The C and FEI+C bars are almost exactly as high as their random-init twins.

Is it real?

- **FEI+C vs released ViT-M** (the planned comparison): $\Delta +0.024$, CI $[-0.022, 0.074]$, $p = 0.147$. **A tie:** FEI+C is at least as good as ViT-M, but not shown to be better.
- **FEI vs released ViT-M** (not pre-declared): $\Delta +0.034$, CI $[-0.007, 0.077]$, $p = 0.052$, higher on 4/5 subsets. Borderline; because this comparison was chosen after seeing the results, treat it as a hint, not a claim.
- **FEI vs rand-FEI**: $\Delta +0.059$, CI $[0.021, 0.098]$, $p = 0.002$. **Significant:** FEI's NMT pre-training carries over to TUAB.
- **C vs rand-C**: $\Delta -0.005$, CI $[-0.035, 0.023]$, $p = 0.642$. **FEI+C vs rand-FEI+C**: $\Delta +0.001$, $p = 0.455$. No gain from pre-training.
- FEI's operating point on the full train set: BAcc 0.785, sensitivity 0.730, specificity 0.840.

Takeaway. FEI+C is healthy and transfers to a new hospital: it ties the released ViT-M and beats band power. Inside FEI+C, the pre-training that transfers is FEI's. C helps on its home data (Experiment 5), but off-domain a random-init C does just as well, so C's value on TUAB comes from its spectrogram input and architecture, not from what it learned.

Limits. TUAB is internal-only (licence). Only 3 random-init seeds. The FEI result is post-hoc. The C and FEI+C encoders saw unlabelled TUAB training data during pre-training, while FEI did not.

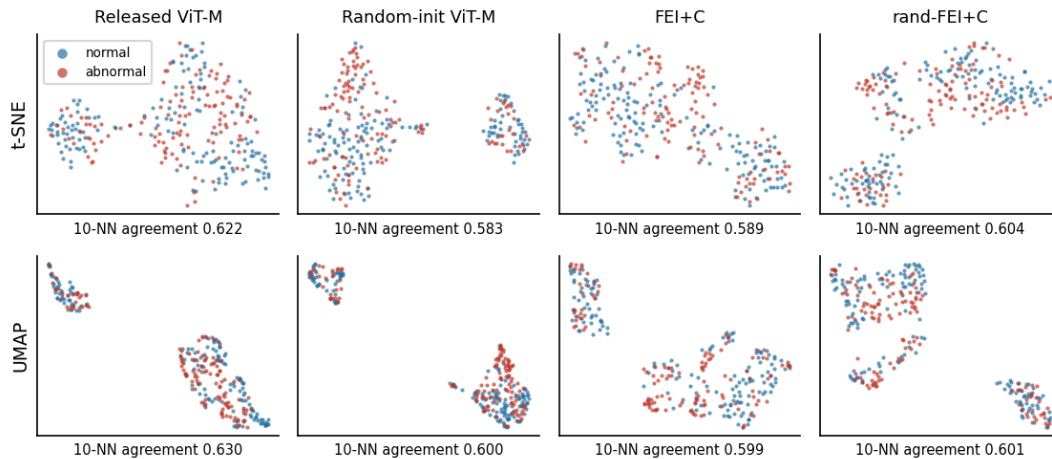
Visual check: t-SNE and UMAP maps

Question. The proposal promised 2-D maps of the learned features, and the paper's Figure 4 shows a UMAP in which normal and abnormal recordings form separate groups. Do our saved TUAB features show that?

Setup.

Item	Value
Recordings	TUAB eval set: 276 recordings (126 abnormal); coloured by the task label only
Features	The saved embeddings from Experiments 4 and 6, one encoder each: released ViT-M, random-init ViT-M (seed 0), FEI+C (fold-0 encoder), rand-FEI+C (seed 0)
Scaling	StandardScaler, the same as the linear probe
t-SNE	scikit-learn, perplexity 30, PCA initialisation, random_state 0
UMAP	umap-learn, 15 neighbours, min_dist 0.1, random_state 0
Check number	10-NN agreement: for each recording, the share of its 10 nearest neighbours that have the same label, averaged. Computed on the original features and on each map. Chance = 0.502

Result.



Each dot = one TUAB eval recording (blue = normal, red = abnormal). Top row t-SNE, bottom row UMAP. The number under each map is its 10-NN label agreement (chance 0.502).

Features (TUAB eval, one encoder each)	Dims	10-NN agreement: original features	t-SNE map	UMAP map
Released ViT-M	768	0.623	0.622	0.630
Random-init ViT-M	768	0.600	0.583	0.600
FEI+C	512	0.629	0.589	0.599
rand-FEI+C	512	0.598	0.604	0.601

How to read it.

- No map shows a normal group and an abnormal group. Blue and red are mixed everywhere.
- Every map, including both random-init ones, splits into separate islands, and every island holds both labels. The islands therefore come from some property of the recordings or their preparation, not from the diagnosis. We did not identify that property.
- The 10-NN agreement is about 0.6 for all four feature sets, against 0.502 for chance. Neighbourhoods are only weakly organised by the label, even though a linear probe on the same features reaches an AUROC of 0.838 (ViT-M, Experiment 4) and 0.862 (FEI+C, Experiment 6). The label is carried by a direction that the probe finds, not by the overall layout that a 2-D map draws.
- The released ViT-M is collapsed (Experiment 3: token std 0.0051 vs 0.634 for random weights), yet its standardised map looks just as structured as the others. Standardising stretches its tiny differences back up to full size. A map therefore cannot reveal collapse; the collapse metrics of Experiment 3 are the test for that.

Is it real? The maps and agreement numbers describe the saved features exactly; they are not a test. On the original features, learned weights score slightly above their random twins (0.629 vs 0.598 for FEI+C, 0.623 vs 0.600 for ViT-M). We did not test these gaps, so they are not a claim.

Takeaway. The proposal's t-SNE / UMAP deliverable is done. On TUAB, none of the maps shows the class groups of the paper's Figure 4, for the released checkpoint or for FEI+C, and random weights give maps that look just as organised. The classification results come from the probes of Experiments 4 and 6, not from these pictures.

Limits. One encoder and one seed per feature set (Experiment 6 averages five encoders). Layouts depend on the map settings, which the paper does not publish. The paper also colours by age and sex; we use the task label only. The stroke cohort (15 subjects) is too small for maps.

Part C — Healthy vs stroke

The proposal's Part 2: classify healthy controls vs stroke patients (Zenodo 19599466) with frozen encoders plus the Brain Symmetry Index (BSI), a clinical measure of left-right power asymmetry. This part reports the clinical baseline (BSI) and our FEI+C.

Experiment 7: pipeline gate and the BSI baseline

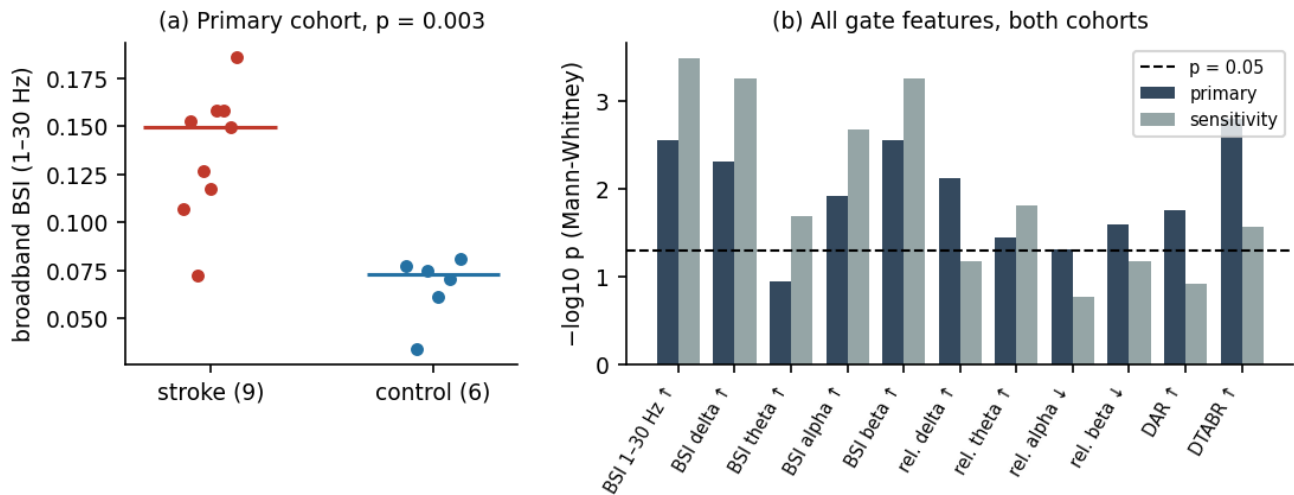
Question. (a) Is our stroke data pipeline sound: does it show the known clinical effect? (b) How well does the clinical BSI measure alone separate stroke patients from controls under the pre-registered leave-one-subject-out test? This is the reference any learned model has to match.

Setup.

Item	Value
Primary cohort	9 stroke patients vs 6 healthy controls, fixed 285 s resting block
Sensitivity cohort	All 18 pre-rehabilitation recordings (10 stroke, 8 control), first 300 s
Gate (pre-declared)	Broadband BSI (1–30 Hz) higher in stroke, Mann-Whitney $p < 0.05$
BSI features	Pairwise-derived BSI over 6 central left/right channel pairs, broadband + 4 bands (5 numbers)
Classifier / CV	StandardScaler + logistic regression ($C = 1$, balanced), leave-one-subject-out; threshold 0.5
Statistics	Bootstrap CI over subjects; 1,000-permutation p

(a) Gate result: PASS in both cohorts (primary $p = 0.003$, sensitivity $p < 0.001$). 10 of 11 classical features differ significantly in the primary cohort, in the expected directions.

Feature	Stroke vs control	Primary: median stroke / control	Primary p	Sensitivity p
BSI 1–30 Hz	↑	0.149 / 0.073	0.003	< 0.001
BSI delta	↑	0.109 / 0.081	0.005	< 0.001
BSI theta	↑	0.098 / 0.062	0.113	0.021
BSI alpha	↑	0.188 / 0.091	0.012	0.002
BSI beta	↑	0.142 / 0.066	0.003	< 0.001
rel. delta	↑	0.485 / 0.214	0.008	0.068
rel. theta	↑	0.172 / 0.106	0.036	0.016
rel. alpha	↓	0.119 / 0.264	0.050	0.173
rel. beta	↓	0.133 / 0.378	0.026	0.068
DAR	↑	3.407 / 0.957	0.018	0.122
DTABR	↑	1.758 / 0.517	0.002	0.027



(a) Broadband BSI per subject; lines = medians. (b) Significance of every gate feature; bars above the dashed line are significant at $p < 0.05$.

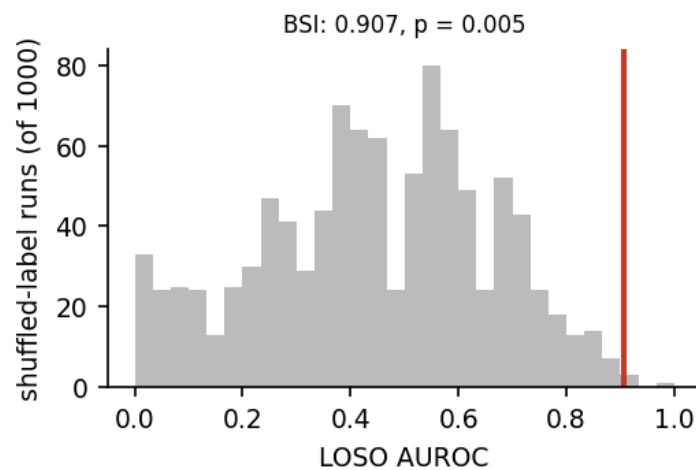
(b) BSI classifier, LOSO.

Primary cohort:

Arm	LOSO AUROC	95 % CI	Permutation p	Sens	Spec
BSI	0.907	[0.679, 1.000]	0.005	0.889	1.000

Sensitivity cohort:

Arm	LOSO AUROC	95 % CI	Permutation p	Sens	Spec
BSI	0.938	[0.775, 1.000]	0.002	0.900	1.000



Permutation test, primary cohort. Grey = AUROC obtained with 1,000 random label shuffles; red line = the real result.

How to read it.

- BSI alone reaches 0.907 on the primary cohort and 0.938 on the sensitivity cohort: strong, and stable when the cohort and recording segment change.
- The shuffled-label AUROCs centre around 0.444, slightly below 0.5, because leave-one-out AUROC is biased a little low with so few subjects. The real result sits far in the right tail.

Is it real? Yes. BSI beats shuffled labels in both cohorts: permutation $p = 0.005$ (primary) and 0.002 (sensitivity). The CIs ([0.679, 1.000] and [0.775, 1.000]) are wide because of the small cohorts.

Takeaway. The pipeline is sound: the classical stroke effect is clearly there, and BSI alone is a strong, stable classifier. It is the bar for FEI+C in Experiment 8.

Limits. Only 15 (18) subjects, so every CI is wide.

Experiment 8: FEI+C on stroke with an NMT-matched input

Question. Our FEI+C encoders were pre-trained on NMT. When the stroke recordings are prepared so that their spectrum matches NMT's, how does FEI+C compare with BSI and with its random-init twin? And are the results robust to the choice of cross-validation?

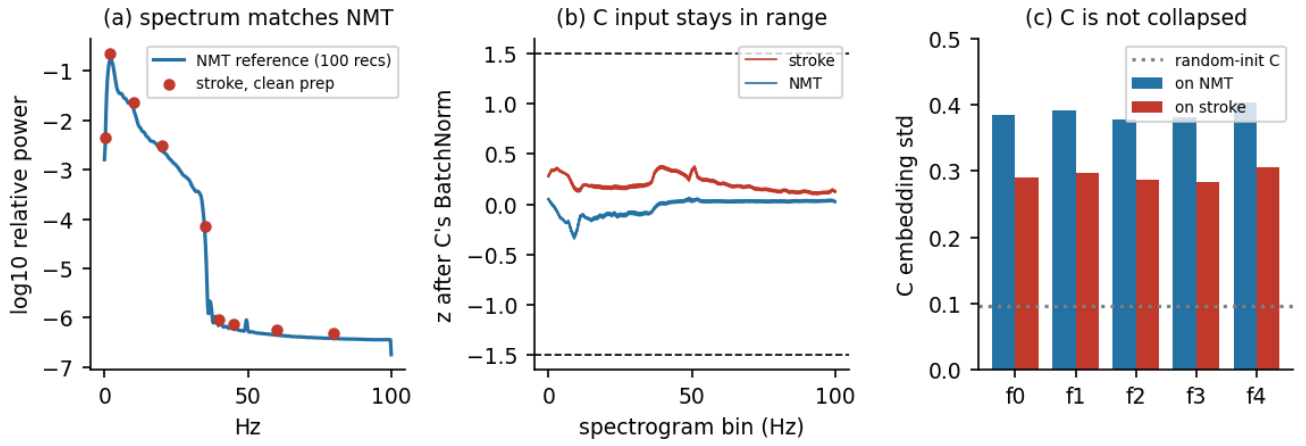
The input preparation for FEI+C was corrected to match NMT's spectrum before this analysis (post-hoc).

Setup.

Item	Value
Cohort	Primary (9 stroke vs 6 control)
FEI+C input prep	Bad-channel interpolation → average reference → 19 channels → 200 Hz, no band-pass → z-score → one fixed, label-free, per-channel filter per cohort that maps the cohort's mean spectrum onto the NMT training mean (100 recordings) → z-score → 2.5 s frames. Design fixed in writing before any run
Encoders	The 5 NMT-only FEI+C fold encoders (unchanged); random-init FEI+C (3 seeds)
Classifier	Same as Experiment 7
CV schemes	LOSO (with 1,000-permutation p), 5-fold ×100 (same 100 splits for every arm), leave-pair-out
Reference	BSI predictions from Experiment 7

Input gates (thresholds fixed in advance). These check that the stroke input now looks like NMT to the encoders:

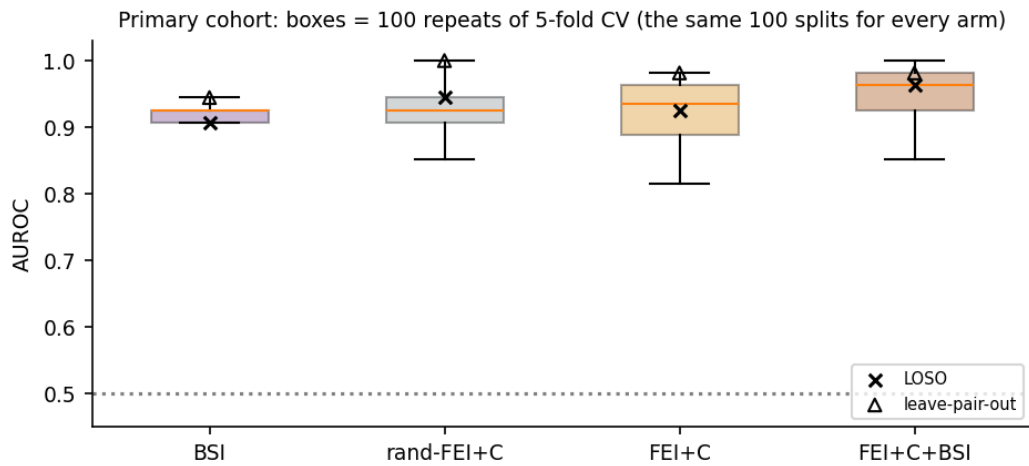
Check (clean prep, primary cohort)	Value	Threshold (fixed in advance)	Result
Largest gap between stroke and NMT spectrum, 0.5–99 Hz (log10 units)	0.184	≤ 0.25	PASS
Largest Branch-C input z-score after its NMT BatchNorm	0.38	≤ 1.5	PASS
C embedding spread on stroke ÷ spread on NMT (worst fold)	0.74	≥ 0.5	PASS
Constant C embedding dimensions	0 %	≤ 5 %	PASS



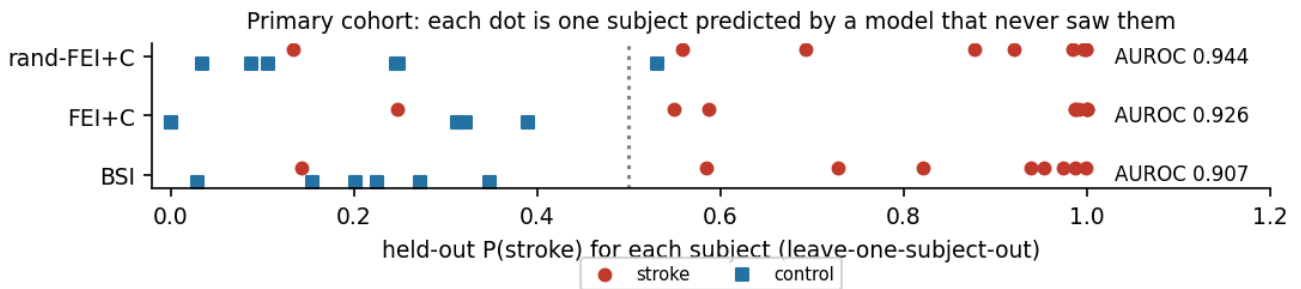
(a) Stroke spectrum after the clean prep (dots) on the NMT reference (line). (b) Branch C's input after its NMT BatchNorm; dashed lines = the ± 1.5 limit. (c) C's embedding spread on NMT and on stroke for each fold encoder; dotted line = random-init C.

Result.

Arm	LOSO	5-fold $\times 100$: mean $\pm$ sd [2.5–97.5 %]	Leave-pair-out	Permutation p (LOSO)
BSI	0.907	0.920 $\pm$ 0.012 [0.907, 0.944]	0.944	0.005
rand-FEI+C	0.944	0.930 $\pm$ 0.037 [0.852, 1.000]	1.000	0.004
FEI+C	0.926	0.918 $\pm$ 0.051 [0.815, 0.981]	0.981	0.003
FEI+C+BSI	0.963	0.951 $\pm$ 0.039 [0.861, 1.000]	0.981	—



Each box = 100 repeats of 5-fold CV; $\times$ = LOSO; Δ = leave-pair-out.



Held-out P(stroke) for each subject under LOSO. Circles = stroke, squares = control; the dotted line is the 0.5 threshold.

FEI+C operating point (LOSO, threshold 0.5): Acc 0.933, BAcc 0.944, Sens 0.889, Spec 1.000 (TP 8, FP 0, TN 6, FN 1).

Paired comparisons:

Comparison (A vs B)	LOSO Δ [95 % CI], $p(\Delta \leq 0)$	5-fold $\times 100$ Δ (A better in % of repeats)	Leave-pair-out Δ [95 % CI], $p(\Delta \leq 0)$
FEI+C vs rand-FEI+C	-0.019 [-0.182, 0.100], $p = 0.826$	-0.012 (41 %)	-0.019 [-0.111, 0.000], $p = 1.000$
FEI+C vs BSI	+0.019 [-0.240, 0.280], $p = 0.523$	-0.001 (60 %)	+0.037 [-0.074, 0.185], $p = 0.421$
FEI+C+BSI vs BSI	+0.056 [-0.120, 0.296], $p = 0.410$	+0.031 (82 %)	+0.037 [-0.074, 0.185], $p = 0.421$

How to read it.

- All gates pass: the stroke spectrum follows NMT's closely, C's inputs stay well inside range (max 0.38, NMT itself 0.34), and C's embeddings are spread out (0.28–0.31 vs 0.38–0.40 on NMT and 0.095–0.098 for random weights), so C is not collapsed.
- FEI+C reaches 0.926 (LOSO), permutation $p = 0.003$. It misclassifies one stroke subject and no control.
- FEI+C and BSI are level under every CV scheme; FEI+C+BSI is higher than BSI in 82 % of the 100 repeats.

Is it real?

- **FEI+C vs BSI:** $\Delta +0.019$, CI [-0.240, 0.280] (LOSO); better in 60 % of repeats. **FEI+C matches BSI.**
- **FEI+C vs rand-FEI+C:** $\Delta -0.019$, CI [-0.182, 0.100]; better in 41 % of repeats. **A tie:** on stroke, FEI+C's pre-training shows no measurable gain as a whole.

Which branch carries the pre-training? (exploratory)

Branch (exploratory)	LOSO	5-fold $\times 100$ mean $\pm$ sd	Leave-pair-out
FEI	1.000	1.000 $\pm$ 0.000	1.000
rand-FEI	0.907	0.903 $\pm$ 0.049	0.981
C	0.833	0.819 $\pm$ 0.063	0.852
rand-C	0.852	0.876 $\pm$ 0.035	0.907

- FEI beats rand-FEI in 100 % of the 5-fold repeats ($\Delta +0.097$). FEI's score is at the 1.0 ceiling with only 15 subjects in a post-hoc arm, so this is not a claim. The leave-pair-out test cannot separate them ($p = 0.558$) because rand-FEI is also near the ceiling.
- C does not beat rand-C (better in 16 % of repeats), the same pattern as on TUAB (Experiment 6).

Takeaway. With an input that matches its training data, FEI+C is healthy on stroke and **matches the clinical BSI baseline**. As on TUAB, the pre-training that carries over is FEI's, not C's; as a whole, FEI+C ties its random-init twin.

Limits. The prep correction is post-hoc (designed and written down before running, but after the first results). Primary cohort only. 15 subjects: all CIs are wide and ceiling effects are possible. "% of repeats" re-splits the same 15 subjects, so it shows robustness to the split, not significance.

Conclusions

Proposal aims and what happened.

Proposal aim	Status	Evidence
Part 1: reproduce the paper's frozen TUAB results with the released checkpoints	Done faithfully; result not reproduced	Pipeline 43/48 files identical to upstream; ViT-M 0.671 ± 0.020 vs paper 0.877 ; best released number 0.838 (Exps 1, 4)
Explain the shortfall	Achieved	Released checkpoints are collapsed on TUAB and stroke (Exp 3); they lose to or tie random weights (Exps 1, 4)
Added: fine-tuning the released ViT-M	Done; no gain from pre-training	Fine-tune 0.755 ± 0.053 vs scratch 0.768 ± 0.114 , $p = 0.81$ (Exp 2)
Changed: FEI+C as a working pre-trained encoder	Achieved	Healthy on NMT (0.865 ± 0.024 , Exp 5); transfers to TUAB and ties ViT-M (0.862 vs 0.838 , Exp 6); FEI's pre-training transfers ($p = 0.002$)
Part 2: healthy vs stroke with frozen encoders + BSI, LOSO	Done	Gate passed ($p = 0.003$); BSI 0.907 (Exp 7); FEI+C 0.926 matches BSI (Exp 8)
Deliverable: t-SNE / UMAP of learned features	Done	No class groups in any TUAB map, learned or random; 10-NN agreement 0.629 for FEI+C vs chance 0.502 (Visual check)
Our own EEG-VJEPa pre-training	Dropped	Out of scope by team decision (2026-09-26)

What we can claim.

1. The released EEG-VJEPa checkpoints do not reproduce the paper's TUAB results, and the reason is representation collapse, not our pipeline.
2. FEI+C is a healthy pre-trained encoder that transfers across hospitals: on TUAB it ties the released ViT-M and beats band power, and its FEI branch's pre-training significantly beats random weights.
3. On stroke, FEI+C matches the clinical BSI baseline.

What we cannot claim. That FEI+C beats ViT-M on TUAB (a tie); that FEI+C's pre-training helps on stroke (it ties random weights); that any model beats BSI on stroke (no comparison is significant); any external benchmark (TUAB numbers are internal; the stroke data set has no published classification benchmark to compare against).

Overall limits.

- **TUAB licence.** Our copy is unlicensed; TUAB numbers are for internal use only.
- **Small stroke cohort.** 15 (primary) and 18 (sensitivity) subjects; every stroke CI is wide.
- **Demographic confound.** Per the Zenodo 19599466 record description, the stroke patients (55.9 ± 14.2 years, 6 male / 4 female) are older and more often male than the controls (46.3 ± 23.6 years, 1 male / 6 female). Encoders can read age and sex from EEG, so part of any stroke-vs-control score may reflect these differences rather than stroke.
- **Post-hoc elements.** The clean stroke prep (Exp 8) and the FEI-vs-ViT-M comparison (Exp 6) were decided after first results.
- **Few seeds.** Random-init controls use 3 seeds; the NMT configuration (Exp 5) has no random-init control.
- **Unpublished details.** The paper does not specify its subsets, reported epoch or fine-tune recipe.

Appendix: where the numbers come from

All numbers are produced by `docs/act0_results/make_results.py`, which reads only saved outputs and asserts that its re-computations match the stored summaries. Main sources (under `code/runs/`):

Experiment	Saved outputs
1	<code>tuab_cached/c_vit{m,b}_{frozen,attentive}_s*</code> , <code>c_vitm_rand*_s*</code> (per-epoch CSV + final predictions)
2	<code>tuab_finetune/{ft,scratch}_vitm_s*_bf16</code>
3	<code>act0/tuab_vitm_linear/results.json</code> + eval features; <code>act0/phaseB/collapse_primary.json</code> + embeddings
4	<code>act0/tuab_vitm_linear/</code> , <code>act0/p13_bandpower/</code>
5	<code>act0/a0_feic_nmt/</code>
6	<code>act0/tuab_feic/</code> (FEI, rand-FEI), <code>act0/tuab_feic_joint/</code> (C, FEI+C and their random controls)
7	<code>act0/phaseB/</code> (gate, LOSO, permutations)
8	<code>act0/phaseB_clean/</code> , <code>act0/phaseB_cv/</code>
V	<code>act0/tuab_vitm_linear/feats_{released,rand0}_eval.npz</code> , <code>act0/tuab_feic_joint/emb*_{f0,rand0}.npy</code>